

Project- proposal



Project Title

Object-Oriented Food Delivery System

M.Mustagees Shahid **(24P-3052)**

Mudassar Iqbal **(24P-3083)**

Supervisor Name & Signature

Dr. Saad Rashad

**FAST-National University of Computer & Emerging
Sciences, Peshawar**

1. Introduction:

With the rise of online food delivery services, a well-structured and efficient system is essential to meet customer demands. This project aims to develop a **Food Delivery System** using **Object-Oriented Programming (OOP)** principles, enabling users to browse restaurants, place orders, and track deliveries seamlessly.

2. Project Goals and Objectives:

- **Design an efficient food ordering system** where users can browse menus and place orders.
- **Implement a restaurant management module** for adding, updating, and removing food items.
- **Enable order tracking functionality** so customers can monitor delivery status.
- **Integrate payment processing options** such as cash on delivery or online payment (if extended to databases).
- **Ensure proper OOP principles** are applied for modular and reusable code.

3. Project Description:

This system will consist of **three main user roles**:

- **Customer:** Can browse menus, place orders, and track delivery status.
- **Restaurant Owner:** Can manage food items, view orders, and update statuses.
- **Delivery Person:** Can update delivery status upon successful completion.

4. Technologies & Tools Used:

- **Programming Language:** C++
- **OOP Concepts:** Encapsulation, Inheritance, Polymorphism, Abstraction.
- **File Handling:** Storing user data, menu items, and order history.
- **Arrays:** Used for managing lists of orders, menus, and users and use of vectors where we need.
- **GUI Framework:** Qt for building an interactive user interface.

5. Expected Outcomes:

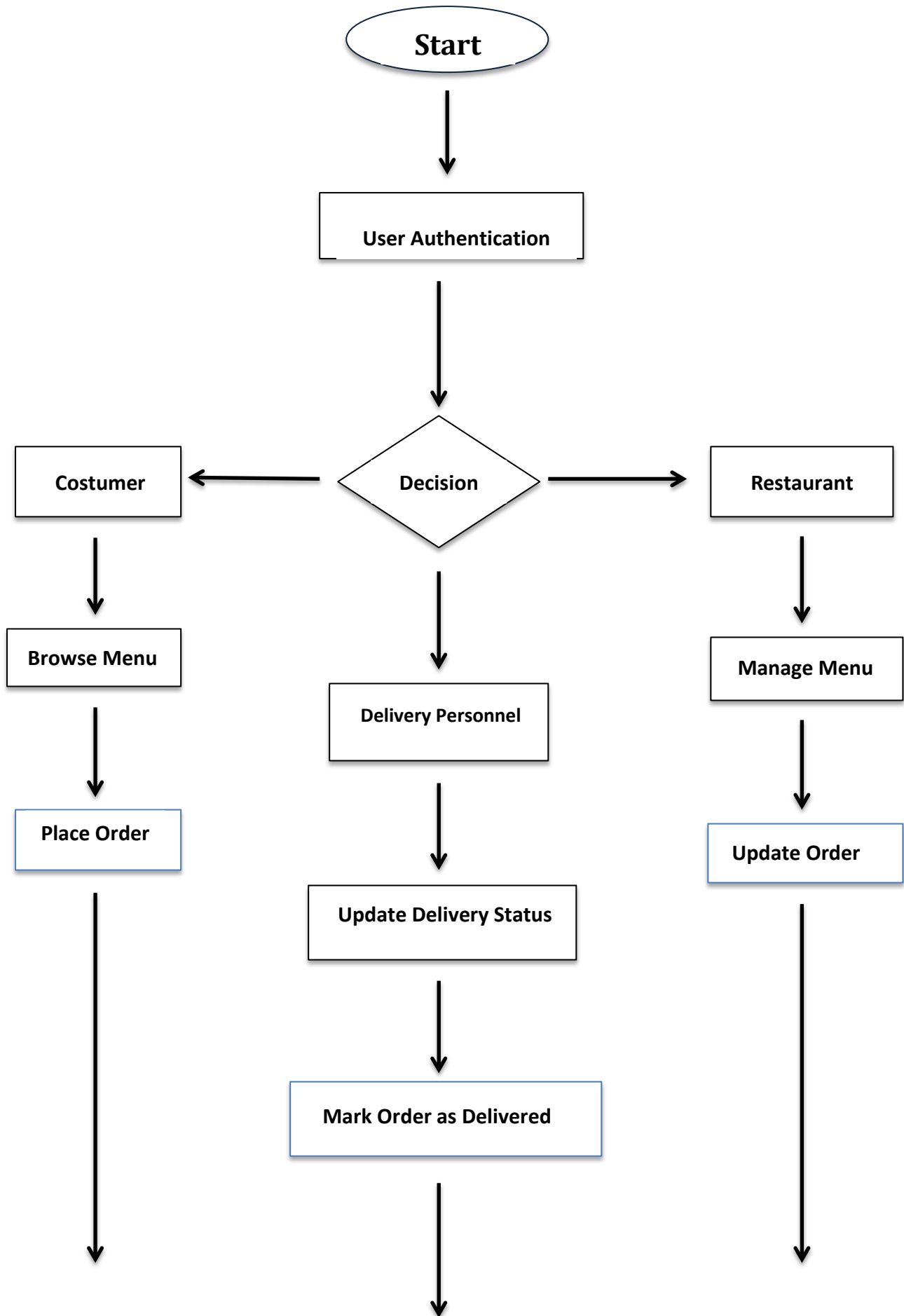
Short-Term Outcomes:

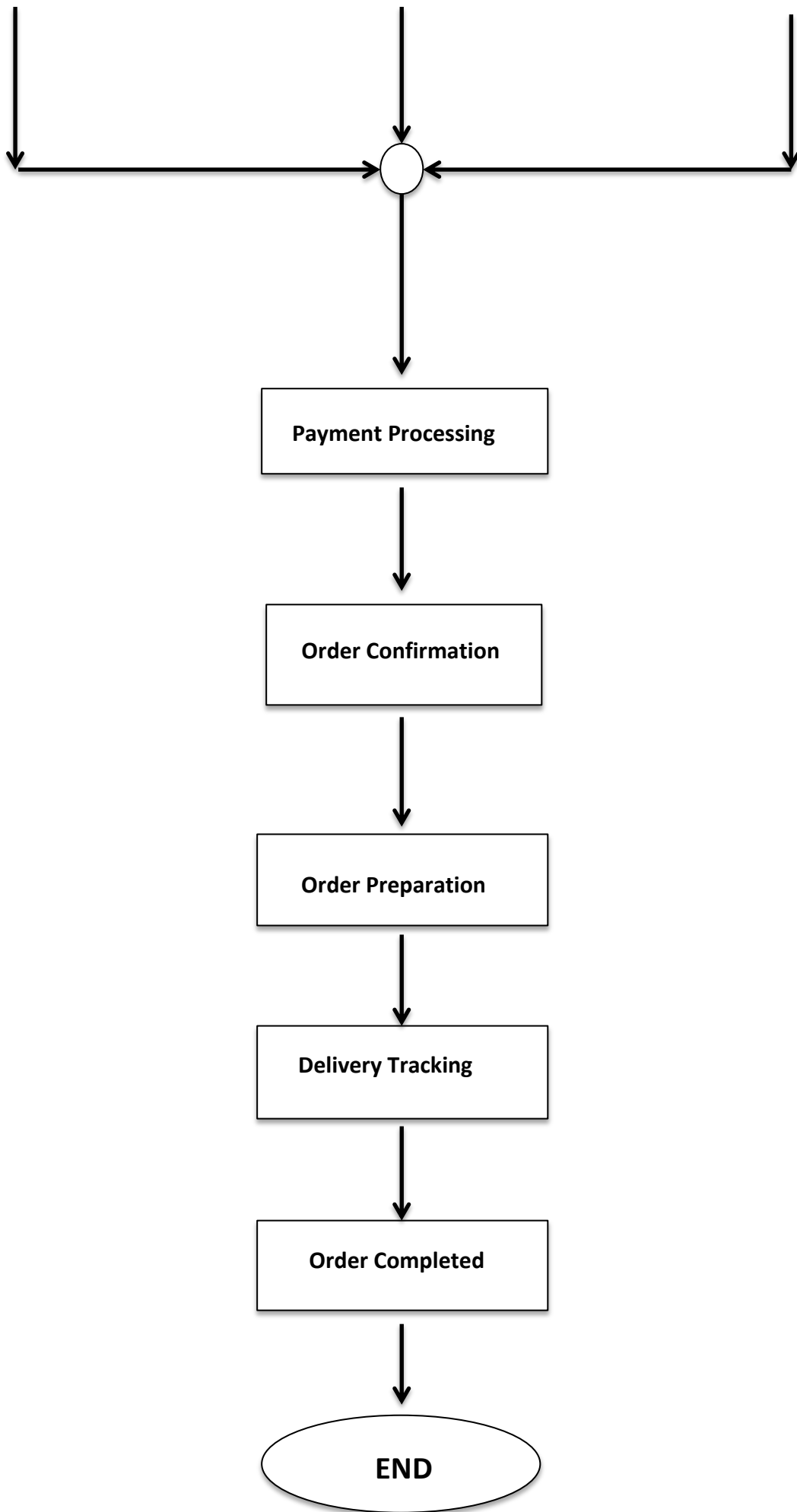
- A basic working system for ordering food and tracking deliveries.
- Implementation of core **OOP concepts** like classes, inheritance, and file handling.
- Successfully tested main functions such as placing and updating orders.

Long-Term Outcomes:

- The system can be improved with more features like **online payments and GPS tracking**.
- A foundation for further research into **AI-based food recommendations**.
- The project can be extended into a **real-world web or mobile application**

FLOW-CHART:





Class-Diagram :

